Artificial Intelligence

Assignment 2

Group: Ritchie

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1. Modeling in first-order logic

Formalize the statements below in first-order logic. Use only the following predicate symbols and equality for this

professor (of arity 1)
confused (of arity 1)
studentOf (of arity 2)

1. Peter is Alice's student. Alice is also someone's student

$\exists x$: studentOf(Alice, x) $\wedge$ studentOf(Peter, Alice)

2. Every person with a student is a professor

$\forall x$ studentOf(_ , x) -> professor(x)

3.Every professor is confused

$\forall x$ professor(x) -> confused(x)

4. Anyone who is not a confused cannot be a professor

$\forall$ x ¬confused(x) -> ¬professor(x)

5. No confused person is both a professor and anyone's student

$\forall$ x [¬confused(x) -> professor(x) $\wedge$ studentOf(x, y)]

6. There is a confused person who is not a professor

$\exists x$: confused(x) $\wedge$ ¬professor(x)

7. Peter and Mary are students of the same professor.

$\exists x$: [studentOf(Peter, x) $\wedge$ studentOf(Mary, x) $\wedge$ professor(x)]

8. People who are confused are not students of a non-confused professor

$\forall$ x confused(x) -> [¬studentOf(x, y) $\wedge$ ¬confused(professor(y))]

9. A student is confused if, and only if, they have a confused professor

$\exists x$ [studentOf(x, y) $\wedge$ confused(x)] $\leftrightarrow$ confused(professor(y))

10. Peter has at least two confused professors

$\exists$ P1,P2 confused(P1) $\wedge$ professor(P1) $\wedge$ confused(P2) $\wedge$ professor(P2) $\wedge$ studentOf(Alice,P1) $\wedge$ studentOf(Alice,P2) $\wedge$ (P1 $\neq$ P2)

(or)

$\exists$ P1,P2 confused(professor(P1)) $\wedge$ confused(professor(P2)) $\wedge$ studentOf(Alice,P1) $\wedge$ studentOf(Alice,P2) $\wedge$ (P1 $\neq$ P2)

2. Reasoning in first -order logic

Consider the following two formulas F, G in first-order logic over the signature of the first exercise:

$$F = \forall X\, (confused(X) \Rightarrow \exists\, Y(confused(Y) \wedge student\ Of(X, Y)))$$

$$G = \forall X(\exists Y\, ((confused(X) \wedge confused\ (Y)) \Rightarrow student\ Of\ (X, Y)))$$

Show that these formulas are not equivalent by finding a model of G that is not a model of F

Answer:

Let's assume a domain consisting of the following elements:

- X1, X2: Individuals representing two different Students
- Y1, Y2: Individuals representing two different Professors.

Consider the following interpretations.

confused(X1) is True
confused(X2) is True
confused (Y1) is True
confused (Y2) is False

studentOf(X1, Y1) is True
studentOf(X2, Y2) is False

We will try to evaluate the formulas F & G.
The formula F states that,

$$F = \forall X\, (confused(X) \Rightarrow \exists\, Y(confused(Y) \wedge student\ Of(X, Y)))$$

Its states for every X ($\forall X$) .
For X = X1 , we have interpretations,

confused(X1) = True

According to the interpretation, there exists Y = Y1 ($\exists$ Y), such that

(confused(Y1) $\wedge$ student Of(X1, Y1))
(T $\wedge$ T)
T

For X = X2 , we have interpretations,

confused(X2) = True

However there is no Y2 , such that

(confused(Y2) $\wedge$ studentOf(X2, Y2)) is True, as

studentOf(X2, Y2) = False

(confused(Y2) $\wedge$ studentOf(X2, Y2)) = False

The formula G states that,

$$G = \forall X(\exists Y ((confused(X) \wedge confused (Y)) \Rightarrow student\ Of (X, Y)))$$

For X = X1 , There exists Y1 we have interpretations,

(confused(X1) $\wedge$ confused (Y1)) $\Rightarrow$ student Of (X1, Y1)

$\neg$(confused(X1) $\wedge$ confused (Y1)) $\vee$ student Of (X1, Y1) is True as

student Of (X1, Y1) is True

For X = X2 , There exists Y2 such that

(confused(X2) $\wedge$ confused (Y2)) $\Rightarrow$ student Of (X2, Y2)

$\neg$(confused(X2) $\wedge$ confused (Y2)) $\vee$ student Of (X2, Y2) is True as

Even student Of (X2, Y2) is False, $\neg$(confused(X2) $\wedge$ confused (Y2)) will be True

Therefore, the formula G is satisfied in the interpretation for all values of X, where ‘F’ is not satisfied in the interpretation for the value of X = X2.

Hence the above is the interpretation of model G but not a model of F, which demonstrates that both models are not equal.

3. Getting familiar with Prolog

In this exercise, we extend the family example from the lecture slides with two new relationships:

1. Siblings have a parent in common. (That technically includes half siblings).
2. An aunt is a female sibling of a parent.

(a) Extend the lecture's rule base by rules for two new prolog predicates siblingof/2 and aunt0f/2 that capture these definitions.

The /2 after the predicate name indicates the arity of the predicate to be developed. You are allowed to use all the predicates from the lecture in your definitions, as well as the build-in predicate X = Y stating that the instantiations of X and Y must be different.

(b) Now we extend the lecture's data base by:

- childOf (caroline, berta) .
- childOf (carl, berta) .
- childOf (michael, claudia).
- female(caroline).

State the result of the query - auntOf (X, Y)

Hint: Part (b) should actually be something that you anyways do voluntarily during part (a): When developing your rules, asking Prolog to show you all assignments that satisfy siblingOf (X,Y) and auntOf(X, Y), respectively, will help you find bugs in your definitions and improve your solution

CODE :-

```
childOf(maria, claudia) .
childOf(claudia, berta) .
childOf(caroline, berta) .
childOf(carl, berta) .
childOf(michael, claudia) .
sibilingOf(berta, claudia) .

female(maria) .
female(claudia) .
female(berta) .
female(caroline) .
grandchild(X, Z) :- childOf(X, Y) , childOf( Y, Z) .
```

```
mother( X) :- female(X), childOf(_,X) .
male(X) :- not(female(X)) .
sibilingOf(X,Y) :- childOf(X,Z), childOf(Y,Z).
auntOf(X,Y) :- female(Y), childOf(X,Z), sibilingOf(Z,Y) .
```

QUERIES :-

```
?- sibilingOf(caroline, claudia) .
true.

?- sibilingOf(caroline, carl) .
true.

?- sibilingOf(caroline, michael) .
false.

?- auntOf(caroline, claudia ) .
true .
```